NoSQL Goals

Goals

- Scalability
- Cost
- Flexibility
- Availability

Scalability

Ability to handle varying workloads

Vertical scaling:

Bigger server

- Expensive
- Bounded
- Simple

Horizontal scaling:

Add more servers

- Cheaper
- Unbounded
- Complex



Cost

RDMBS (Oracle)

- Expensive
- Pay for support
- Huge set of add-ons

NoSQL

- Free / Open source
- Pay for support
- Large set of adds-ons

Aside: Why not dump Oracle?

Technical:	Performance about 2X open-source
Business:	Customers use related business tools
Cost:	Switching difficult

Flexibility

RDMBS

Model:

- Flexible
- Significant data modeling
- Mostly static structure

Language:

- SQL

NoSQL

Model:

- Flexible
- Minimal data modeling
- Dynamic structure

Language:

- Flexible

Availability

RDMBS

- Single system
  - Single point of failure
- Single database
- Should replicate data

NoSQL

- Distributed system
  - Fault-tolerant
- Often multiple data sources
- Should replicate data

Summary of characteristics

- Distributed
- Asynchronous
- Performance > consistency
  - Reads > writes
  - Analysis > transactions
- Data model more(?) flexible